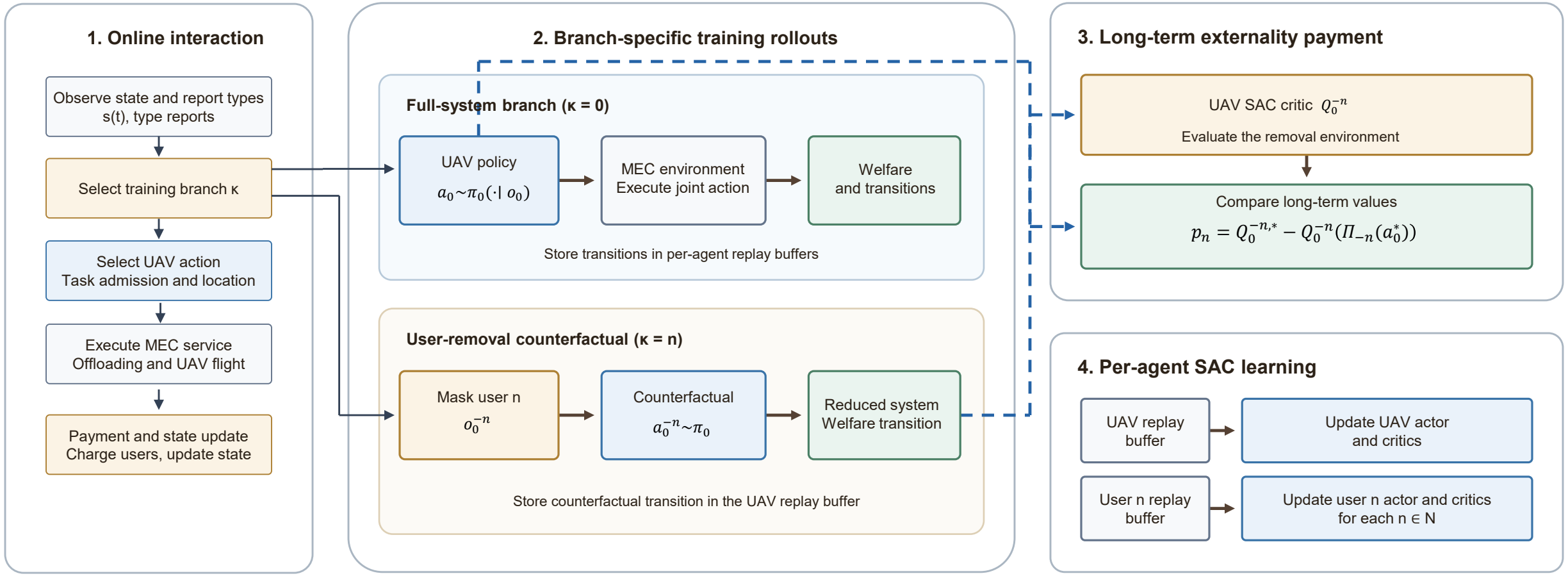

1. Online interaction
Observe state and report types
s(t), type reports
Select training branch κ
Select UAV action
Task admission and location
Execute MEC service
Offloading and UAV flight
Payment and state update
Charge users, update state
2. Branch-specific training rollouts
Full-system branch (κ = 0)
UAV policy
$a_0 \sim \pi_0(\cdot \mid o_0)$
MEC environment
Execute joint action
Welfare
and transitions
Store transitions in per-agent replay buffers
User-removal counterfactual (κ = n)
Mask user n
o_0^{-n}
Counterfactual
$a_0^{-n} \sim \pi_0$
Reduced system
Welfare transition
Store counterfactual transition in the UAV replay buffer
3. Long-term externality payment
UAV SAC critic Q_0^{-n}
Evaluate the removal environment
Compare long-term values
$p_n = Q_0^{-n,*} - Q_0^{-n}(\Pi_{-n}(a_0^*))$
4. Per-agent SAC learning
UAV replay
buffer
Update UAV actor
and critics
User n replay
buffer
Update user n actor and critics
for each n ∈ N